* Precipitation reactions :

Solubility Guidelines for Ionic Compounds

Solubility : the amount of the substance that can be dissolved in a given quantity of solvent at the given temp.

Any substance with a solubility less than 0.01 mol/L will be considered insoluble.

SOLUBLE IONIC COMPOUNDS

1- Compounds of alkali metal group (1A) and of the ammonium (NH_4^+) are soluble in water

2- nitrates NO_3^-, acetates CH_3COO^-

3- Cl^-, Br^-, I^- containing compounds except compounds of Ag^+, Pb^{2+}, Hg_2^{2+}

4- SO_4^{2-} containing compounds except compounds of Sr^{2+}, Ba^{2+}, Pb^{2+}, Hg_2^{2+}

INSOLUBLE IONIC COMPOUNDS

1- S^{2-}, OH^- containing compounds
except of alkali metal (1A)
NH_4^+
Ca^{2+}, Sr^{2+}, Ba^{2+}

2- CO_3^{2-}, PO_4^{3-} containing compounds
except compounds of alkali metal (1A)
NH_4^+

For exmp. $(NH_4)_2CO_3$ is soluble

H_2CO_3 is soluble
Ag_3PO_4 is insoluble

Example : $CuSO_4(aq) + Na_2CO_3(aq) \longrightarrow CuCO_3(s)\downarrow + Na_2SO_4(aq)$
precipitates — soluble

$Mg(NO_3)_2(aq) + NaOH(aq)$ write net ionic equation and determine precipitation.

$\rightarrow Mg(OH)_2(s)\downarrow + NaNO_3(aq)$

$Mg^{2+}(aq) + \cancel{NO_3^-(aq)} + \cancel{Na^+(aq)} + OH^-(aq) \longrightarrow Mg(OH)_2(s) + \cancel{Na^+(aq)} + \cancel{NO_3^-(aq)}$

$Mg^{2+}(aq) + OH^-(aq) \longrightarrow Mg(OH)_2(s)$

* Classify the following compounds as soluble or insoluble in water :

$Co(OH)_2$ → insoluble (in 1)

$Ba(NO_3)_2$ → soluble (sol 2)

$(NH_4)_3(PO_4)$ → soluble (in 2)

$PbSO_4$ → insoluble (sol 4)

* Write balanced net ionic equations, identify spectator ions.

$Ba^{+2}(NO_3)_2(aq) + K_2SO_4^{-2}(aq) \longrightarrow BaSO_4(s)\downarrow + (NO_3)_2SO_4(aq)$

$Fe(NO_3)_2(aq) + 2KOH(aq) \longrightarrow Fe(OH)_2(s)\downarrow + 2KNO_3(aq)$

$Ba^{2+} + 2(NO_3^{-1}) + K^{+1} + SO_4^{2-}(aq) \longrightarrow BaSO_4(s) + K^{+1}(aq) + 2NO_3^{-1}$

spectator ions : K^{+1}
NO_3^{-1}

$Fe^{2+}_{(aq)} + 2NO_3^{-1}(aq) + 2K^{+}_{(aq)} + 2(OH^-)_{(aq)} \longrightarrow Fe(OH)_2(s)\downarrow + 2K^{+}(aq) + 2(NO_3^-)_{(aq)}$

$Fe^{2+}(aq) + 2(OH^-)(aq) \longrightarrow Fe(OH)_2(s)\downarrow$ net ionic equation

spectator ions : K^{+1}
NO_3^-

Oxidation – Reduction Reactions

Oxidation number: the electron number which an atom gives or takes when become compound

Rules for determining oxidation numbers

(If two rules contradict with each other, use above rule to determine.)

1- For free element atoms (did not form a compound with other atoms)
Oxidation number: 0
Cl ⟶ 0
Al ⟶ 0

2- a) For nötr species summation of oxidation number: 0
$[CH_3OH]$ ⟶ 0
$[MgCl_2]$ ⟶ 0

b) For an ion oxidation number is equal to charge of ion
Fe^{3+} ⟶ +3
MnO_4^- ⟶ -1

3- Group 1 metals have oxidation number +1

Group 2 metals have oxidation number +2 at their compounds

KCl $\longrightarrow$ K^{+1}

K_2CO_3 $\longrightarrow$ K^{+1}

$MgBr_2$ and $Mg(NO_3)_2$ $\longrightarrow$ Mg^{+2}

4- Fluorine at its compounds oxidation number is -1

HF $\longrightarrow$ F^{-1}

SF_6 $\longrightarrow$ F^{-1}

5- Hydrogen at its compounds oxidation number +1

CH_4 $\longrightarrow$ H^{+1}

NH_3 $\longrightarrow$ H^{+1}

H_2S $\longrightarrow$ H^{+1}

6- Oxygen at its compounds oxidation number -2

H_2O, CO_2

7- For binary compounds of metals

with group 7 elements → oxidation number -1

with group 6 elements → oxidation number -2

With group 5 elements → oxidation number -3

$MgBr_2 \longrightarrow Br^{-1}$

$Li_2S \longrightarrow S^{-2}$

$Li_3N \longrightarrow N^{-3}$

Find the oxidation number of underlined elements.

P_4 → Rule 1, P ox. number: 0

Al_2O_3 → Rule 2, summation of ox. number: 0
Rule 6, O ox. number: -2
Al ox. number: +3

MnO_4^- → Rule 2b, summation of ox. number of an ion is charge of the ion: -1
O ox. number: -2
Mn ox. number: +7

NaH → Rule 3, Na ox. Number: +1
Rule 5, H ox. Number: +1
if both of them +1 then $(NaH)^{2+}$! → This contradicts with Rule 2a
Rule 2 and Rule 3 is above Rule 5, so
Na ox. number: +1
H ox. number: -1

Step 1: Write half redox reactions
Step 2: For each half reaction, balance atom numbers except H and O
Step 3: For each half reaction, balance O atom by adding H_2O
Step 4: For each half reaction, balance H atom by adding H^+ (acidic media)
Step 5: Balance e^- numbers between two half reactions
Write summation of two half reactions

Write balanced equations for following redox reactions performed in acidic solutions.

$$MnO_4^- + I^- \rightarrow Mn^{2+} + I_2$$

MnO_4^- ⟶ O^{-2}

Mn^{+7}

Oxidation: $2\,I^- \rightarrow {I_2}^0 + 2e^-$

Reduction: $MnO_4^- + 5e^- \rightarrow Mn^{+2}$

Step 2 completed

Oxidation: $2\,I^- \rightarrow {I_2}^0 + 2e^-$

Reduction: $MnO_4^- + 5e^- + 8H^+ \rightarrow Mn^{+2} + 4H_2O$

Step 4 completed

Oxidation: $2\ I^- \rightarrow\ {I_2}^0 + 2e^-$ $\times 5$

Reduction: $+\quad MnO_4{}^- + 5e^- + 8H^+ \rightarrow\ Mn^{+2} + 4H_2O$ $\times 2$

$$10\ I^- + 2\ MnO_4{}^- + 16\ H^+ \rightarrow 5\ I_2 + 2\ Mn^{+2} + 8\ H_2O$$

$$BrO_3{}^-{}_{(aq)} + N_2H_{4(l)} \rightarrow Br^-{}_{(aq)} + N_{2(g)}$$

$N_2H_4 \longrightarrow H^{+1}$, N^{-2}

$BrO_3{}^- \longrightarrow O^{-2}$, Br^{+5}

Oxidation: $N_2H_{4(l)} \rightarrow N_2{}^0{}_{(g)} + 4e^- + 4H^+{}_{(aq)}$

Reduction: $BrO_3{}^-{}_{(aq)} + 6e^- \rightarrow Br^-$

Step 2 completed

Oxidation: $N_2H_{4(l)} \rightarrow N_2{}^0{}_{(g)} + 4e^- + 4H^+{}_{(aq)}$

Reduction: $BrO_3{}^-{}_{(aq)} + 6e^- + 6H^+ \rightarrow Br^- + 3\,H_2O$

Step 4 completed

Oxidation: $N_2H_{4(l)} \rightarrow {N_2}^{0}{}_{(g)} + 4e^- + 4H^+{}_{(aq)}$ $\times 3$

Reduction: $+ \quad {BrO_3}^-{}_{(aq)} + 6e^- + 6H^+ \rightarrow Br^- + 3\,H_2O$ $\times 2$

$$3N_2H_{4(l)} + 2{BrO_3}^-{}_{(aq)} \rightarrow 2N_{2(g)} + 2\,Br^-{}_{(aq)} + 6H_2O_{(l)}$$

Write balanced equations for following redox reactions performed in basic solutions.

$$MnO_4^-{}_{(aq)} + CN^-{}_{(aq)} \rightarrow MnO_2{}_{(s)} + OCN^-{}_{(aq)}$$

MnO_4^- → O^{-2}, Mn^{+7} $\qquad$ MnO_2 → Mn^{+4}

CN^- → N^{-3}, C^{+2} $\qquad$ OCN^- → N^{-3}, O^{-2}, C^{+4}

Oxidation: $CN^-{}_{(aq)} + H_2O_{(l)} \rightarrow OCN^-{}_{(aq)} + 2e^- + 2H^+{}_{(aq)}$

Reduction: $MnO_4^-{}_{(aq)} + 3e^- + 4H^+{}_{(aq)} \rightarrow MnO_2{}_{(s)} + 2H_2O_{(l)}$

Write half reactions with adding
H_2O and H^+ if needed

Oxidation: $CN^-_{(aq)} + H_2O_{(l)} \rightarrow OCN^-_{(aq)} + 2e^- + 2H^+_{(aq)}$ $\times 3$

Reduction: $+ \quad MnO_4^-{}_{(aq)} + 3e^- + 4H^+_{(aq)} \rightarrow MnO_{2(s)} + 2H_2O_{(l)}$ $\times 2$

Balance e- numbers
Write summation of two half reactions, remove H^+ and H_2O to simplfy

$$3\,CN^-_{(aq)} + 2MnO_4^-{}_{(aq)} + 2H^+_{(aq)} \rightarrow 3\,OCN^-_{(aq)} + 2\,MnO_{2(s)} + H_2O_{(l)}$$

Add 2 OH^- to both sides of equation for converting the acidic media to basic

$$3\,CN^-_{(aq)} + 2MnO_4^-{}_{(aq)} + 2H^+_{(aq)} + 2OH^- \rightarrow 3\,OCN^-_{(aq)} + 2\,MnO_{2(s)} + H_2O_{(l)} + 2OH^-$$

$$3\,CN^-_{(aq)} + 2MnO_4^-{}_{(aq)} + 2H_2O_{(l)} \rightarrow 3\,OCN^-_{(aq)} + 2\,MnO_{2(s)} + H_2O_{(l)} + 2OH^-_{(aq)}$$

$$3\,CN^-_{(aq)} + 2MnO_4^-{}_{(aq)} + H_2O_{(l)} \rightarrow 3\,OCN^-_{(aq)} + 2\,MnO_{2(s)} + 2OH^-_{(aq)}$$

Activity Series of Metals in Aqueous Solutions

TABLE 4.5 • Activity Series of Metals in Aqueous Solution

Metal	Oxidation Reaction
Lithium	$Li(s) \longrightarrow Li^+(aq) + e^-$
Potassium	$K(s) \longrightarrow K^+(aq) + e^-$
Barium	$Ba(s) \longrightarrow Ba^{2+}(aq) + 2e^-$
Calcium	$Ca(s) \longrightarrow Ca^{2+}(aq) + 2e^-$
Sodium	$Na(s) \longrightarrow Na^+(aq) + e^-$
Magnesium	$Mg(s) \longrightarrow Mg^{2+}(aq) + 2e^-$
Aluminum	$Al(s) \longrightarrow Al^{3+}(aq) + 3e^-$
Manganese	$Mn(s) \longrightarrow Mn^{2+}(aq) + 2e^-$
Zinc	$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$
Chromium	$Cr(s) \longrightarrow Cr^{3+}(aq) + 3e^-$
Iron	$Fe(s) \longrightarrow Fe^{2+}(aq) + 2e^-$
Cobalt	$Co(s) \longrightarrow Co^{2+}(aq) + 2e^-$
Nickel	$Ni(s) \longrightarrow Ni^{2+}(aq) + 2e^-$
Tin	$Sn(s) \longrightarrow Sn^{2+}(aq) + 2e^-$
Lead	$Pb(s) \longrightarrow Pb^{2+}(aq) + 2e^-$
Hydrogen	$H_2(g) \longrightarrow 2\,H^+(aq) + 2e^-$
Copper	$Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$
Silver	$Ag(s) \longrightarrow Ag^+(aq) + e^-$
Mercury	$Hg(l) \longrightarrow Hg^{2+}(aq) + 2e^-$
Platinum	$Pt(s) \longrightarrow Pt^{2+}(aq) + 2e^-$
Gold	$Au(s) \longrightarrow Au^{3+}(aq) + 3e^-$

Ease of oxidation increases

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$Zn_{(s)}$ is oxidized by Cu^{+2} aq. Solution but $Ag_{(s)}$ is not.

$$Zn^0{}_{(s)} + Cu^{+2}{}_{(aq)} \rightarrow Zn^{+2}{}_{(aq)} + Cu^0{}_{(s)}$$

$$Ag^0{}_{(s)} + Cu^{+2}{}_{(aq)} \rightarrow reaction\ does\ not\ occur$$

$$Cu^0{}_{(s)} + 2\,Ag^{+1}{}_{(aq)} \rightarrow Cu^{+2}{}_{(aq)} + 2\,Ag^0{}_{(s)}$$

Any metal on the list can be oxidized by the ions of elements below it.

* Stochiometry of Aqueous Solution's Reactions

Calculate the molarity of a solution made by dissolving 23.4 g Na_2SO_4 in water to form 125 mL of solution?

$$\frac{23.4\ g\ Na_2SO_4}{125\ mL} \times \frac{1\ mol\ Na_2SO_4}{142.1\ g\ Na_2SO_4} \times \frac{1000\ mL}{1\ L} = 1.32\ M$$

How many grams of Na_2SO_4 are required to make 0.35 L of 0.5 M Na_2SO_4?

$$\frac{n\ mol\ Na_2SO_4}{0.35\ L} = 0.5\ M \Longrightarrow 0.175\ mol\ Na_2SO_4 \text{ required}$$

$$0.175\ mol\ Na_2SO_4 \times \frac{142.1\ g}{1\ mol} = 24.9\ gram\ Na_2SO_4$$

How many mL of 3.0 M H_2SO_4 are needed to make 450 mL of 0.1 M H_2SO_4 ?

$$M_c \cdot V_c = M_d \cdot V_d$$

$$3\ M \times V_c = 0.1\ M \times 450\ mL$$

$$V_c = 15\ mL$$

How many grams of $Ca(OH)_2$ are needed to neutralize 25.0 mL of 0.1 M HNO_3 ?

Find moles of HNO_3 $\quad \frac{n\ mol\ HNO_3}{0.025\ L} = 0.1\ M \Rightarrow 0.0025\ mol\ HNO_3$

$2\,HNO_3 + Ca(OH)_2 \longrightarrow Ca(NO_3)_{2(aq)} + 2\,H_2O_{(l)}$ Balanced eq

0.0025 mol $\quad$ 0.00125 mol $\times$ 74.1 g/mol $= \boxed{0.0926\ g\ Ca(OH)_2}$

A sample of 70.5 mg of K_3PO_4 is added to 15 mL of 0.05 M $AgNO_3$, resulting in the formation of a precipitate.

a) write molecular eq. for reaction

b) what is the limiting reactant

c) Calculate the theoretical yield, in grams, of the precipitate.

a) $\underbrace{K_3PO_4(aq)}_{1} + \underbrace{3\,AgNO_3(aq)}_{3} \longrightarrow 3KNO_3(aq) + Ag_3PO_4\downarrow(s)$

0.000332 0.00075

b) 0.000332 → 0.000986 → limiting reactant $AgNO_3$

$$\frac{n \text{ mol } AgNO_3}{0.015\ L} = 0.05 \quad n = 0.00075 \text{ mol } AgNO_3$$

$$70.5 \text{ mg } K_3PO_4 \times \frac{10^{-3}\ g}{1\ mg} \times \frac{1 \text{ mol}}{212.3\ g\ K_3PO_4} = 0.000332 \text{ mol } AgNO_3.$$

c) $\underbrace{K_3PO_4(aq)}_{0.00025} + \underbrace{3AgNO_3(aq)}_{0.00075} \longrightarrow 3KNO_3(aq) + Ag_3PO_4\downarrow(s)$

↓

+ 0.00025 mol Ag_3PO_4

× 418.7 g/mol = 0.1 g Ag_3PO_4